There Is No Post Hurricane World

Data Mining to Uncover Persistent Effects of Natural Disasters

TEAM

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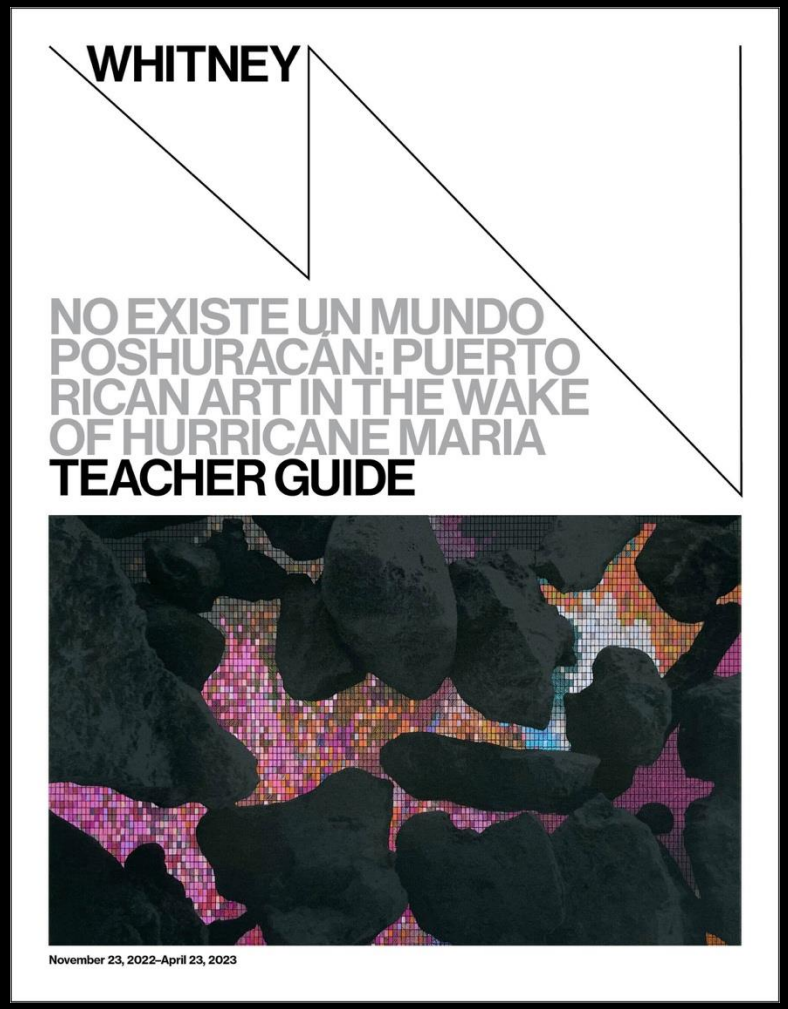
QUESTIONS



INITIAL:

1. Is Puerto Rico truly trapped in the wake of Hurricane Maria?
2. Are other countries trapped in the wake of severe natural disasters?
3. What conditions produce the persistent effects of natural disasters?
4. What methods can we use to study the persistent effect of natural disasters?

FOLLOWING EDA:

1. Which areas of social and economic development were permanently affected by Hurricane Maria?

Datasets

1. DISASTER DATASET

+ Selections from EM-DAT, the International Disaster Database
+ https://www.kaggle.com/datasets/jseebs/disaster-dataset

2. ECONOMIC AND SOCIAL DATA

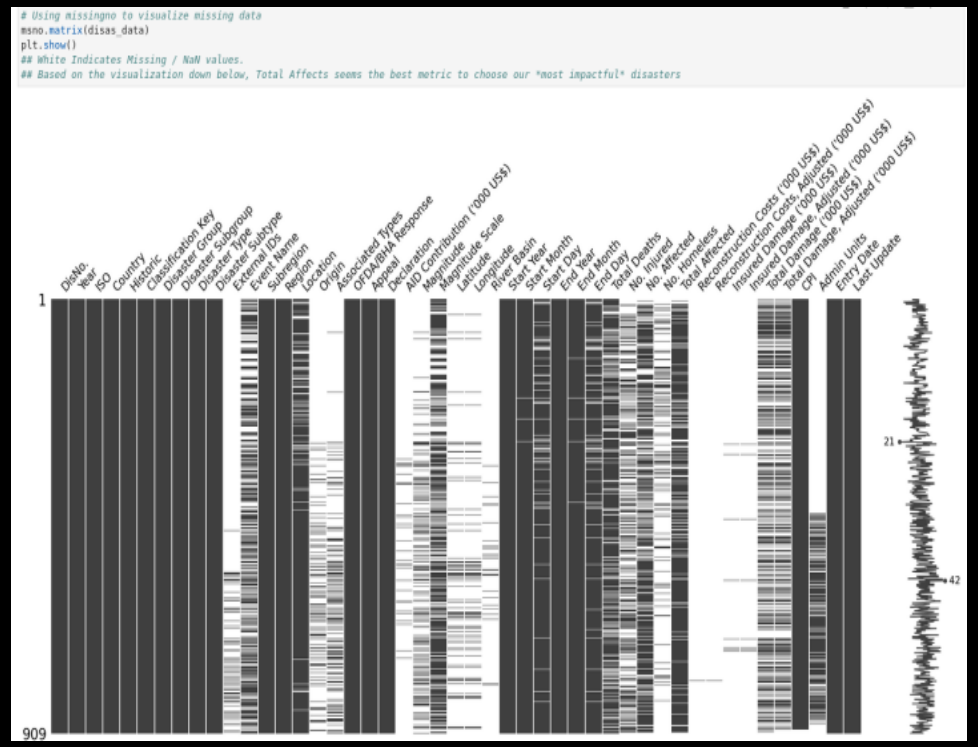
+ All selections, for all countries, in the WDI World Bank's World Development Indicators Database
+ https://databank.worldbank.org/

List of Tools

- **Programming Language:** Python
- **Data Analytics:** Pandas, NumPy, SciPy, Matplotlib, Seaborn
- **Collaboration:** Github, Jupyter
- **Database Management:** World Bank Group API, CSV files

Exploratory Data Analysis

1. Determine thresholds of significance
2. Select 100 randomized social and economic Development Indicators (“Indicators”) in 4 countries: the Philippines, Puerto Rico, Japan, and the United States of America.
3. We then conducted a time-series analysis between 1960 and 2023 for these Indicators by plotting the data and marking years with natural disasters in the selected countries.
4. We failed to find answers to our Initial questions -- except in Puerto Rico.



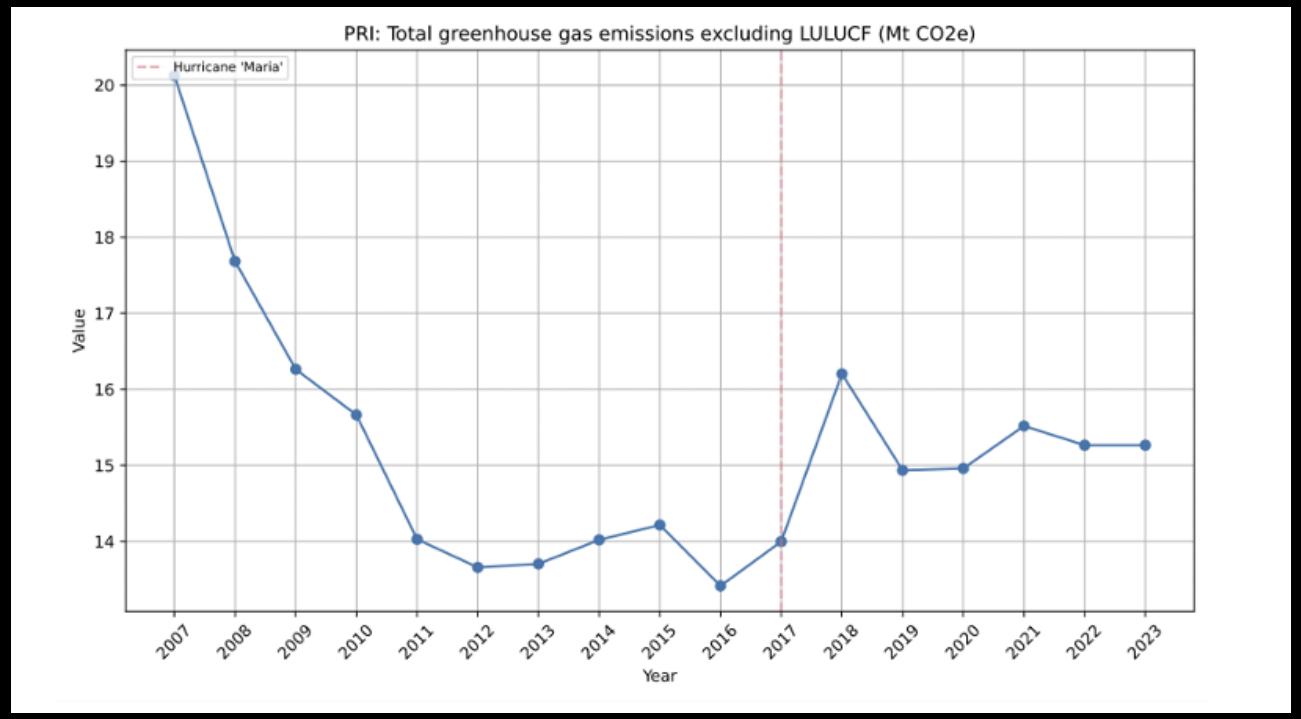
Mining Techniques: Data Pipeline

1. EXTRACT
   + Query World Development Indicator API
   + Pandas to extract Disaster data from EM-DAT csv file
2. TRANSFORM
   + Eliminating NULL Values
   + Back/Front Fill and Linear Interpolation to handle missing values
   + Reducing dimensions to significant storms
3. LOAD
   + Into DataFrames
4. TIME-SERIES ANALYSIS
   + Plot Development Indicators, mark Hurricane Maria

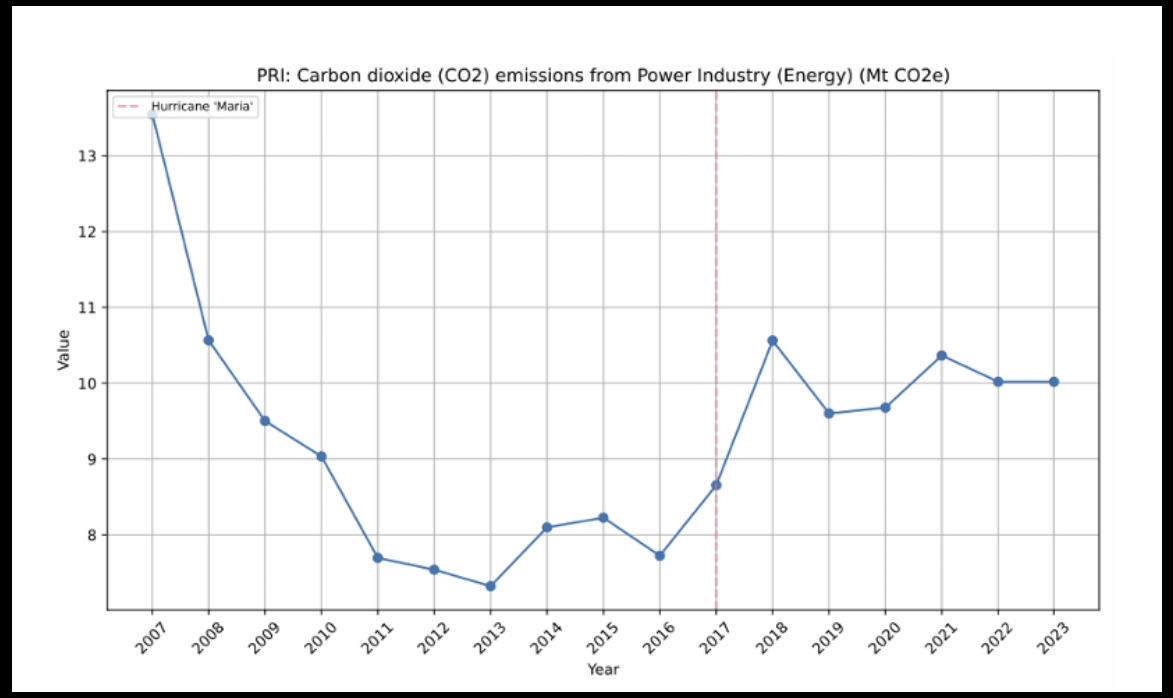
Key Findings

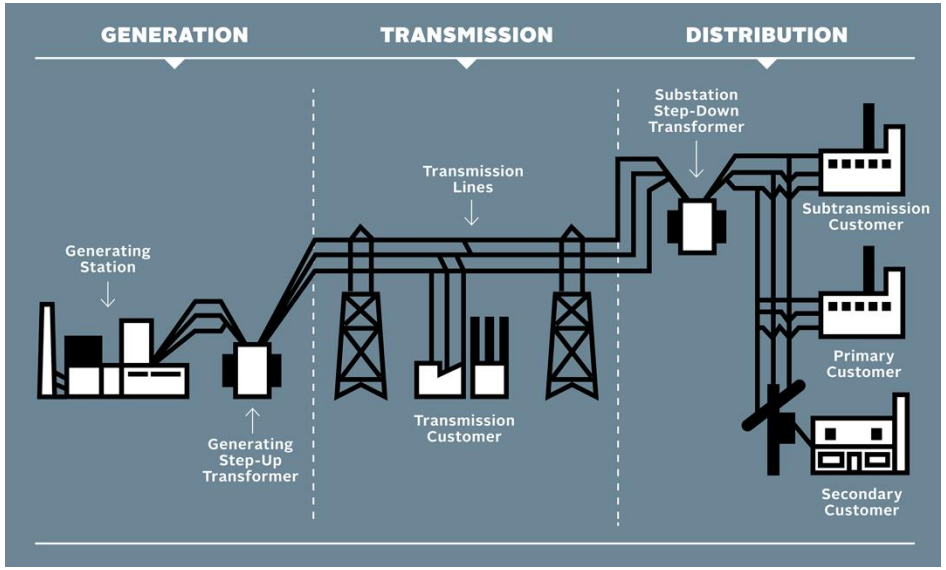
Maria negatively and permanently affected the following areas of socioeconomic development in the Commonwealth:

1. Food Production and Nutrition
2. Public Sentiment
3. Energy
4. Education
5. Economic Productivity



EXAMPLE PIPELINE OUTPUT: ENERGY



Applying Findings

We believe that the dynamics outlined in previous sections compound each other, and that unraveling a decades-long tangled web of problems requires closing the biggest wound first. Puerto Rico must fix its outdated, dirty, and inefficient power grid. After all, many of the struggles we documented trace back to the historic blackout that followed Maria's arrival to Puerto Rico.